

OXYPC INVENTORY LIFECYCLE

PRODUCT

Product Requirements (PRDs)

Requirements for the top improvement opportunities, plus the template for new ones.

Last updated: 25 Oct 2026 · Source: staging branch @ 55ce58a

App Development Team · Bluemonk

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In this document: Part A — PRD-01 · Part Request & PNA Status · Part B — PRD-02 · Fail Reasons & Rework Routing · Part C — PRD-03 · Grading-Driven Pricing · Part D — PRD-04 · Invoice & E-way Bill from Sale · Part E — PRD Template

PART A

PRD-01 · Part Request & PNA Status

Field	Value
Status	Draft
Roadmap item	N2 (Now, Q4 2026) — 02_Roadmap.md
Business owner	Spare Parts Manager (with Repair supervisor)
Personas	Repair Engineer (L1/L2, L3/L4), Spare Parts Manager, Production Manager, TRC Manager, Sales Manager (closes sourcing)
Related	Proposed KPIs P3 PNA wait time, P4 Parts fill rate (05); Roadmap X3 Harvested parts first

1. Problem

An engineer who needs a part raises a request and then has to walk to Stores or open each Device Detail page to learn what happened to it. If the part cannot be found, the engineer marks it "PNA" (Part Not Available) in a different place, and nothing ties that mark to the request. Stores cannot see how long tags have been waiting, and no one is told when a part is handed over, sourced or declared not in stock. Tags sit in L1/L2 with an open PNA and cannot be completed (`routers/repair.py::l1l2_complete_to_stress` blocks Mark Complete while PNA is open), but the reason and age are hard to see in one place.

2. Personas

Persona	Need
Repair Engineer	See every request's status on my queue; be told when my part is ready
Spare Parts Manager	One list of open requests and PNA tags, oldest first, with what is blocking each
Production Manager	Know which tags are stalled on parts, and for how long
TRC Manager	PNA count and age across the floor
Sales Manager	Know which sourcing requests are urgent because tags are waiting

3. Goals and non-goals

Goals	Non-goals
One visible status per part request, on the engineer's queue and Device Detail	Changing how sourcing deals are closed in the CRM
PNA linked to the request it came from (when there is one)	Automatic purchase orders
Notify the engineer on handover / not in stock / sourced	Changing stock-ledger arithmetic
Keep every step's time so wait time and fill rate can be measured	Harvest-first suggestion (Roadmap X3, separate PRD)
Keep PNA history across repeated mark / clear cycles	

4. Current behaviour (from code)

Area	What happens today	Source
Raise request	From Device Detail Parts Consumption (single or multi), L1/L2 Bulk Part Request, L3/L4 Request Parts modal. Duplicate open request (requested / handed_over / procure) for same part is refused in bulk.	`routers/part_requests.py::create_part_request`, `create_part_requests_batch`; `routers/repair.py::bulk_part_request`
Request statuses	`requested`, `handed_over`, `received` ("Part Changed"), `not_in_stock`, `procure` ("Sent for Sourcing"), `scrapped`, `warranty_sent`, `warranty_replaced`, `cancelled`	`models/part_request.py::PartRequest`, `routers/part_requests.py`
Timestamps	One `actioned_at` / `actioned_by` pair, overwritten by every step (handover, verify, not in stock ...). Step times are lost.	`routers/part_requests.py::handover_part`, `validate_receiving`, `not_in_stock`
Spare Parts actions	Handover (qty = requested), Not In Stock, Procure (creates `PartSourcingRequest` `open`), Scrap, Warranty send / receive, Bulk action	`routers/part_requests.py`
Sourcing loop	Sales Manager closes the deal in CRM; Spare Parts Manager verifies → stock credited via ledger; originating request goes back to `requested` (shown as "Available")	`close_sourcing`, `verify_sourcing`
Engineer visibility	Status badges only on Device Detail Parts Consumption	`templates/devices/detail.html` (badges for `not_in_stock`, `procure`, `received`)
Notifications	None in the part-request flow	`routers/part_requests.py` (no `create_notification` call)
PNA mark	Any signed-in user can mark / unmark a part PNA for a tag; one row per (tag, part name); unmark sets `is_active = false`; re-mark reopens the same row, so only the latest cycle is kept; "Remove All PNA" clears all	`models/pna_part.py::DevicePNAPart`, `routers/part_requests.py::mark_part_pna`, `clear_all_pna`
PNA ↔ request	No link. `not_in_stock` does not create PNA; PNA does not change the request.	both models
PNA effect	Blocks L1/L2 Mark Complete; counted on Production "PNA" tile (L1/L2 only); PNA summary CSV (L1/L2/L3); PNA column on WorkID Status; `pna_map` on L3/L4	`routers/repair.py`, `routers/stock.py::trc_production_list`, `trc_production_pna_summary`, `routers/workid_status.py`, `services/pna_lookup.py`

Area	What happens today	Source
Audit	Every step audited (`PART_HANDOVER` , `PART_RECEIVED` , `PART_NOT_IN_STOCK` , `PART_PROCURE` , `PART_PNA_MARKED` , …)	`routers/part_requests.py`

5. User stories and acceptance criteria

US-01 — See part status on my queue

As a Repair Engineer, I want a Parts column on my L1/L2 and L3/L4 queue, so that I know which tags I can work on now.

#	Given	When	Then
AC-01.1	A tag on my L1/L2 queue has 2 open requests (one `handed_over` , one `procure`)	I open `/repair/l1`	The Parts column shows "1 ready to verify · 1 sourcing" with coloured pills
AC-01.2	A tag has an active PNA part	I open my queue	The Parts column shows a red "PNA · n days" pill; hovering lists part names, who marked, when
AC-01.3	A tag has no requests and no PNA	I open my queue	The Parts column shows "—"
AC-01.4	I filter the queue by "Waiting on parts"	I apply the filter	Only tags with an open request or active PNA are listed, and the count matches the Production PNA tile for L1/L2

US-02 — Be told when my part moves

As a Repair Engineer, I want a notification when my part is handed over, declared not in stock or becomes available after sourcing, so that I do not have to chase Stores.

#	Given	When	Then
AC-02.1	I raised a request	Spare Parts Manager clicks Handover	I get an in-app notification "RAM 8GB handed over for — verify on Device Detail" linking to the tag
AC-02.2	I raised a request	Spare Parts Manager clicks Not In Stock	I get a notification, and the request shows "Not In Stock" on my queue
AC-02.3	The request was sent for sourcing	Spare Parts Manager verifies the sourcing	I get "Part now available" and the request shows "Available"

US-03 — Not In Stock creates PNA

As a Spare Parts Manager, I want "Not In Stock" to mark the part PNA on the tag (linked to the request), so that the tag is visibly blocked without a second step.

#	Given	When	Then
AC-03.1	An open request for "RAM" on tag T	I click Not In Stock	A PNA row for (T, "RAM") is active with `source = not_in_stock` and `part_request_id` set; audit `PART_PNA_MARKED` written
AC-03.2	T already has an active PNA for "RAM" (marked by the engineer)	I click Not In Stock	The existing PNA row is linked to the request; no duplicate

#	Given	When	Then
AC-03.3	A linked PNA exists	The part is later handed over	The PNA is cleared automatically (`cleared_by` = Spare Parts Manager) and the history keeps both events

US-04 — One list of what is blocked

As a Spare Parts Manager, I want a "Blocked tags" tab on Part Master listing every active PNA and every open request older than 1 working day, oldest first, so that I work the longest waits first.

#	Given	When	Then
AC-04.1	There are active PNA rows and open requests	I open `/spare-parts` → Blocked tags	A Global Table lists Tag, Stage, Part, Status, Linked request, Sourcing status, Age (days), Engineer, sorted by Age desc
AC-04.2	I tick rows and choose Handover	I submit	Same rules as Bulk action; result alert lists moved and skipped tags

US-05 — Keep the whole PNA story

As a TRC Manager, I want every mark and clear of a PNA kept, so that I can explain a delay even after the part was re-marked.

#	Given	When	Then
AC-05.1	Part was marked, cleared, marked again	I open Device Detail → PNA history	Both cycles are listed with who / when
AC-05.2	The WorkID Status PNA history column	I export	Both cycles are in the export

6. Functional requirements

ID	Requirement	Priority
FR-01	Record a timestamp and user for each step of a part request (requested, handed over, received, not in stock, procure, available, scrapped, cancelled, warranty). Do not overwrite earlier steps.	Must
FR-02	Show a derived **Part status** per request with these labels: Requested · Handed over (verify) · Changed · Not in stock · Sourcing · Available · PNA · Scrapped · Cancelled · Warranty.	Must
FR-03	Add a Parts column and a "Waiting on parts" filter to `/repair/l1` and `/repair/l3l4`.	Must
FR-04	Send an in-app notification (`services/notifications.py::create_notification`) to the requesting engineer on Handover, Not In Stock and Sourcing Verified.	Must
FR-05	Not In Stock creates or links an active PNA row for the same tag and part; Handover or Verify of a linked request clears it.	Must
FR-06	Keep every PNA mark / clear cycle as history (no overwrite on re-mark).	Must
FR-07	Add "Blocked tags" tab on Part Master (Global Table, oldest first, CSV export).	Should
FR-08	Show PNA age (working days) wherever PNA is shown (queues, Production tile tooltip, WorkID Status).	Should

ID	Requirement	Priority
FR-09	Include L3/L4 tags in the Production "PNA" tile (today L1/L2 only) or rename the tile "PNA (L1/L2)".	Should
FR-10	Daily digest notification to Spare Parts Manager: count of requests open > 1 working day and PNA > 5 working days.	Could

7. Data and permission changes (Proposed)

Schema (dbdiagram approval required; additive):

Table	Change	Notes
`part_request_events` (new)	`id`, `part_request_id` FK, `from_status`, `to_status`, `by_username`, `at`, `notes`	Append-only step log. Backfill one row per existing request from `created_at` and current `actioned_at`.
`part_requests`	add `handed_over_at`, `received_at` (nullable)	Fast reads for P4 fill rate; also written to the events table
`device_pna_parts`	add `part_request_id` FK (nullable)	Link PNA ↔ request
`device_pna_events` (new)	`id`, `pna_part_id` FK, `action` (marked / cleared), `by_username`, `at`, `source`	Keeps every cycle; existing row stays the "current state"

Audit actions: reuse existing ones; add `PART_PNA_LINKED` (FR-05) and `PART_PNA_AUTO_CLEARED`.

Permissions: No new module. The Parts column is visible to anyone who can see the repair queue. "Blocked tags" tab follows the existing Part Master view permission. PNA mark stays open to any signed-in user (current rule).

8. UX notes

- Parts column pills: Handed over = `bg-primary`; Changed = `bg-primary` outline; Sourcing = `bg-info text-dark` (as today on Device Detail); Not in stock / PNA = `bg-danger`; Available = success `#0F7A37`.
- PNA pill text "PNA · 6d"; tooltip lists parts, marked by, marked at (same pattern as WorkID Status `pna_tip`).
- "Blocked tags" tab is a Global Table card with Sky Blue header; toolbar holds Bulk Handover / Not In Stock / Export.
- Bulk results use the Design System multi-line `alert-info` ("12 handed over, 2 skipped — already actioned: ...").
- Notifications use the existing bell; type `info` for handover / available, `warning` for not in stock.

9. Metrics

Metric	Baseline	Target	Source
PNA wait time (median, mark → clear)	baseline: to be measured	≤ 5 working days *(proposed)*	`device_pna_events`
Parts fill rate (handover ≤ 1 working day)	baseline: to be measured	≥ 85% *(proposed)*	`part_requests.handed_over_at − created_at`

Metric	Baseline	Target	Source
Tags blocked by PNA (count, daily)	baseline: to be measured	−30% in 3 months *(proposed)*	Production PNA tile
Engineer visits to Device Detail just to check part status	baseline: to be measured (access log)	−50% *(proposed)*	server log

10. Rollout

Phase	Scope	Gate
1	FR-01, FR-02, FR-03, FR-04, FR-06 + backfill	Staging UAT: REP-01…03, SPM-01…02 and new US-01/02/05 scenarios
2	FR-05 (Not In Stock ↔ PNA link)	Spare Parts Manager agrees auto-PNA rule
3	FR-07…FR-10	KPI baselines (Roadmap N6)

Setting: `AppSetting` key `pna_auto_from_not_in_stock` (default on) so FR-05 can be switched off. Manual page update in `utils/manual_content.py` (Part Master, L1/L2 Repair).

11. Open questions

#	Question	Owner
1	Should Not In Stock always create PNA, or only when no sourcing is planned?	Spare Parts Manager
2	Working-day calendar: which days and holidays count?	TRC Manager
3	Should Sales Manager get a notification when a sourcing request is blocking > N tags?	Sales Manager
4	Should engineers be allowed to cancel their own request? Today `cancelled` is set only via bulk action.	Repair supervisor

PART B

PRD-02 · Fail Reasons & Rework Routing

Field	Value
Status	Draft
Roadmap item	N3 phase 1 (Q4 2026) and X6 phase 2 (Q1 2027) — 02_Roadmap.md
Business owner	Final QC lead (with QC lead and Cosmetic Lead)
Personas	QC Inspector (Stress), Cosmetic Lead, Final QC Inspector, Repair Engineer, Production Manager, TRC Manager
Related	Proposed KPI P2 First-pass yield (05); `tests/test_final_qc_fail_assign.py`, `tests/test_cosmetic_fail_button.py`, `tests/test_failed_fqc_fail_note_tooltip.py`

1. Problem

A tag can fail at Stress Test, at four cosmetic steps and at Final QC. Each fail is written into a single "current value" field on the device, and the next event clears or overwrites it. When the tag reaches the next engineer, the earlier fail story is gone from the screen; it survives only as free text in the stage-movement log. Stress and Cosmetic fails always go to L1/L2, while Final QC routes by reason. Engineers repeat diagnosis, repeat fails are invisible, and first-pass yield cannot be measured cleanly.

2. Personas

Persona	Need
Repair Engineer	See every past fail on the tag (what, where, who, when) before I start
QC Inspector	Record a reason category, not only checklist items, so the tag goes to the right place
Cosmetic Lead	Fail a tag back for hardware or for rework inside the paint line
Final QC Inspector	See if this tag failed before and why; route failed tags in bulk
Production Manager / TRC Manager	Count rework loops per tag, per stage, per engineer

3. Goals and non-goals

Goals	Non-goals
Every fail is kept as a structured event on the tag	Changing the Stress Test checklist itself
The open (unresolved) fail is shown on the receiving queue	Changing who can pass / fail (permissions stay)

Goals	Non-goals
One reason-category list and one routing rule for Stress, Cosmetic and Final QC	Automatic engineer scoring or penalties
Rework count visible per tag	Replacing `StageMovement` as the movement log

4. Current behaviour (from code)

Area	What happens today	Source
Stress fail	Fail modal with failed / N/A checklist items + note; engineer must be L1/L2; tag → `l1`, new L1 WorkOrder, notification. `Device.stress_notes = "Failed: <items> — <note>"`. Checklist kept in `StressTestResult` (`results_json`).	`routers/stress_api.py::stress_fail_assign`, `_record_checklist_result`
Stress pass	`stress_notes` set to `None` on Complete to paint	`routers/stress_api.py::stress_complete_to_paint`
Cosmetic fail	Allowed at Cosmetic Received, Cleaning, Water Sanding, Cosmetic Completed; engineer must be L1/L2; tag → `l1`; `Device.repair_notes` **overwritten** with " Failed — assigned to . "	`routers/cosmetic.py::cosmetic_fail_assign`
Other `repair_notes` writes	Overwritten by Request L3/L4 and by L3/L4 Complete	`routers/repair.py::request_l3l4`, `l3l4_complete`
Final QC fail	Reason from Master Data `qc_failure_reason` + required Final Notes; tag → `final_qc_fail_hold`; `fqc_failure_reason`, `fqc_final_notes` set (single value); bucket gets the latest reason / engineer	`routers/cosmetic.py::advance_stage`, `templates/cosmetic/final_qc.html`
Final QC routing	"Move" / "Bulk Move" routes by `FAIL_REASON_TO_DEPT`: Hardware → L1/L2 (to last engineer, else parked in `trc_production`), Software → Stress Test, Cosmetic → Cosmetic Received. Any other Master Data reason is **skipped** ("No recognized Failure Reason").	`routers/cosmetic.py::fqc_move_failed`, `FAIL_REASON_TO_DEPT`, `_resolve_fail_engineer`
Final QC edit	Edit clears `final_qc_status`, `fqc_failure_reason`, `fqc_final_notes`, `fqc_pass_notes` and returns tag to `final_qc`	`routers/cosmetic.py::fqc_edit_revert`
History	Free text in `StageMovement.notes` ("Final QC Failed — ...", " Failed — ...")	`models/device.py::StageMovement`
Where shown	L1/L2 queue: Repair Notes and Stress Notes columns; Devices Failed table: failure reason, final notes; Device Detail notes	`templates/repair/*.html`, `templates/cosmetic/final_qc.html`, `templates/devices/detail.html`
Audit	Cosmetic fail audited (`COSMETIC_FAIL_ASSIGNED`); Stress / Final QC fail rely on StageMovement + WorkOrder	as above

5. User stories and acceptance criteria

US-01 — Fail history on the tag

As a Repair Engineer, I want a Fail history panel on Device Detail and an "open fail" line on my queue, so that I start from what already failed.

#	Given	When	Then
AC-01.1	Tag T failed Stress (items: Keyboard, Wi-Fi) and later Final QC (Hardware: "hinge loose")	I open Device Detail for T	Fail history lists both, newest first: stage, category, items, note, failed by, failed at, routed to, resolved at
AC-01.2	T's latest fail is open	I open `/repair/l1`	T shows "Open fail: Final QC · Hardware · hinge loose" and a "Rework #2" badge
AC-01.3	T passes Stress after the fix	The pass is saved	The Stress fail event gets `resolved_at` / `resolved_by`; Stress Notes column no longer shows it as open, but history keeps it

US-02 — Same reasons everywhere

As a QC Inspector or Cosmetic Lead, I want to pick a reason category (Hardware / Software / Cosmetic / Other) when I fail a tag, so that routing and reports work the same way as Final QC.

#	Given	When	Then
AC-02.1	Fail modal on Stress or Cosmetic	I open it	A required Reason category dropdown appears (Master Data `qc_failure_reason`), plus the existing checklist / note
AC-02.2	I submit without a category	—	The modal keeps my values and shows `alert alert-danger small`: "Choose a fail reason"

US-03 — Route by rule (phase 2)

As a Final QC Inspector, I want every failed tag to route by a rule I can see, so that nothing is skipped because its reason is not one of three hard-coded words.

#	Given	When	Then
AC-03.1	Admin maps each `qc_failure_reason` value to a destination in Master Data	I Bulk Move failed tags	Each tag goes to its mapped destination; the result alert lists moved and skipped tags with reasons
AC-03.2	A reason has no mapping	I Bulk Move	Tag is skipped and the alert says "No route for reason '' — ask Admin to map it"
AC-03.3	Stress fail with category Software	QC Inspector submits	Tag stays at Stress for OS reinstall (no L1 hand-off) *(proposed rule; confirm in Open questions)*

US-04 — See rework loops

As a Production Manager, I want rework count per tag and fail counts per stage and reason, so that I can fix root causes.

#	Given	When	Then
AC-04.1	Fail events exist	I open the Flow KPIs page (Roadmap X5)	First-pass yield per stage and top 5 reasons per stage are shown, matching the events table

6. Functional requirements

ID	Requirement	Priority
FR-01	Write one fail event for every fail at Stress, Cosmetic (all fail points) and Final QC, in the same transaction as the stage move.	Must
FR-02	Mark the open fail event resolved when the tag next passes the stage where it failed (Stress complete-to-paint; cosmetic advance past that stage; Final QC pass).	Must
FR-03	Show Fail history on Device Detail and an "Open fail" + "Rework #n" on L1/L2, L3/L4, Stress, Cosmetic and Final QC queues.	Must
FR-04	Keep `stress_notes`, `repair_notes`, `fqc_failure_reason` working as today (backward compatible) but stop relying on them as history.	Must
FR-05	Reason category required on Stress and Cosmetic fail (Master Data `qc_failure_reason`).	Should (phase 2)
FR-06	Routing map from reason to destination stage held in Master Data / AppSetting, used by Final QC Move and by Stress / Cosmetic fail; replaces hard-coded `FAIL_REASON_TO_DEPT`.	Should (phase 2)
FR-07	Final QC "Edit" (revert decision) marks the fail event `voided`, not deleted.	Must
FR-08	Backfill fail events from `StageMovement.notes` patterns ("Final QC Failed —", "Failed — assigned to", Stress moves to `l1`) and `StressTestResult` FAIL rows.	Should
FR-09	Audit `DEVICE_FAIL_RECORDED`, `DEVICE_FAIL_RESOLVED`, `DEVICE_FAIL_VOIDED`.	Must

7. Data and permission changes (Proposed)

Schema (dbdiagram approval required; additive):

Table	Columns	Notes
`device_fail_events` (new)	`id`, `device_id` FK, `barcode`, `stage` (DeviceStage at fail), `reason_category`, `failed_items` (JSON), `notes`, `routed_to_stage`, `routed_to_user_id`, `failed_by`, `failed_at`, `resolved_at`, `resolved_by`, `status` (open / resolved / voided), `rework_seq`, `stress_result_id` (nullable FK)	Index on (`device_id`, `status`)
Master Data	new category `fail_reason_route` (reason → destination)	Phase 2

Permissions: unchanged. Fail actions keep their gates (`cosmetic_finalqc` edit for Final QC; stage module edit for Cosmetic; Stress page permission). Fail history is read-only for anyone who can open Device Detail.

8. UX notes

- Device Detail: a "Fail history" card (Global Table style, Sky Blue header) under Stage History; open fail row highlighted with `table-warning`; resolved rows show a success `#0F7A37` tick.
- Queues: one compact line under the tag: red badge "Rework #2" + "Final QC · Hardware · hinge loose" (tooltip shows full note, same pattern as the existing Final QC fail-note tooltip).
- Fail modals: Reason category first, then checklist, then note; errors per Design System §09 (modal stays open, values kept).
- Bulk Move result: multi-line `alert-info` with moved and skipped tags in `font-monospace`.

9. Metrics

Metric	Baseline	Target	Source
First-pass yield at Final QC	baseline: to be measured (from StageMovement)	≥ 85% *(proposed)*	`device_fail_events`
First-pass yield at Stress	baseline: to be measured	to be agreed	`device_fail_events`
Tags with ≥ 2 rework loops	baseline: to be measured	−30% in 2 quarters *(proposed)*	`rework_seq`
Final QC fails skipped at Move for unrecognised reason	baseline: to be measured	0	Move results

10. Rollout

Phase	Scope	Gate
1 (Q4 2026)	FR-01…04, FR-07…09; backfill	UAT: QC-02, COS-04, FQC-02, FQC-03 + US-01
2 (Q1 2027)	FR-05, FR-06 routing map	Final QC lead approves the map; UAT US-02, US-03

11. Open questions

#	Question	Owner
1	Should a Stress "Software" fail stay in Stress (OS reinstall) instead of going to L1/L2?	QC lead
2	Should Cosmetic be able to fail back to an earlier cosmetic step (e.g. Painting → Putty) instead of only to L1/L2?	Cosmetic Lead
3	Which Master Data `qc_failure_reason` values exist in production today, beyond Hardware / Software / Cosmetic?	Admin
4	Do we count a Final QC "Edit" revert as a fail for yield? (Proposed: no — voided.)	Final QC lead

PART C

PRD-03 · Grading-Driven Pricing

Field	Value
Status	Draft
Roadmap item	X1 (Next, Q1 2027) — 02_Roadmap.md
Business owner	Sales lead (with Final QC lead and Director)
Personas	Final QC Inspector, Sales Executive, Managers/Admin, Inventory Manager
Related	BRD v6 objective 4 "Grade-driven pricing" (`docs/generate_brd_v6.js`); audit "Commercial 3.5/10" and P-4 `expected_sale_value` (`docs/audits/2026-04-29-claude-md-full-audit.md`); proposed KPI P5 Grade mix

1. Problem

Final QC gives every passed tag a grade, but the grade plays no part in its selling price. Ready to Sale computes a minimum price from cost plus a margin, so two identical laptops — one A grade, one C grade — get the same floor. Sales sets real prices by hand, one tag or one selection at a time. The CRM already holds a Grade Price Matrix, but nothing on the refurbishment side reads it. The BRD has promised grade-driven pricing since v6.

2. Personas

Persona	Need
Final QC Inspector	Know the grade I give changes the price, so grading is consistent
Sales Executive	See a suggested price per tag that reflects grade, and a floor I should not go below
Managers/Admin	Maintain one grade price list; see margin by grade
Inventory Manager	Know expected value of stock by grade

3. Goals and non-goals

Goals	Non-goals
A suggested min / max selling price per tag from grade + device type (+ optional make / generation)	Dynamic market pricing from external sites
Keep the cost-based floor; warn when the grade price is below it	Blocking a sale (stays a warning, as today)
Sales can override with a reason; every change audited	Changing how As-Is lots are priced (phase 2)
One grade scale used by Final QC, Sales and CRM	Re-grading historical sold tags

4. Current behaviour (from code)

Area	What happens today	Source
Grade capture	Final QC Pass form offers A / B / C ("— No change —" keeps the old grade); only saved on Pass	`templates/cosmetic/final_qc.html`, `routers/cosmetic.py::advance_stage`
Grade enum	`DeviceGrade` = A, B, C, D, E, scrap	`models/device.py::DeviceGrade`
CRM grades	`GRADES = A, B, C, D, Mix, As-Is, Scrap`	`models/crm.py`
Cost on pass	Final QC Pass sets `device_price = base + consumed parts + changed parts`, once per tag	`routers/cosmetic.py::advance_stage` ("price finalized")
Ready to Sale min price (auto)	`unit price + other rate (CostConfig cosmetic_rate) + item margin (amount or %)`; unit price = `device_price` or lot buying price ÷ qty	`services/ready_pricing.py::tag_min_price`, `unit_price`, `load_ready_rates`
Manual price	Set price / Bulk set price write `Device.min_selling_price`, `max_selling_price`; audited `TAG_PRICE_SET` / `TAG_PRICE_BULK_SET`; role check `_role_can_set_price`	`routers/sales.py::set_ready_tag_price`, `bulk_set_ready_tag_price`
As-Is lot price	Lot total + as-is margin; stored overrides in `AsIsLotPrice`	`services/ready_pricing.py::as_is_min_price`, `routers/sales.py` As-Is routes
Sale guard	Below-cost warning only (does not block)	`routers/sales.py::create_sale` (`below_cost_warning_for`)
Grade Price Matrix	Table `crm_grade_price_matrix`: device_type, grade, material_type, brand, min/max buy price, `target_sell`, `min_margin_pct` (default 15). Used only by its own CRUD + CSV upload pages.	`models/crm.py::GradePriceMatrix`, `routers/crm_price_matrix.py`
Expected sale value	`expected_sale_value` is never populated, so the auto-scrap / scrap-warning rule never fires	audit Apr-29 P-4; `services/cost_engine.py`

5. User stories and acceptance criteria

US-01 — Suggested price from grade

As a Sales Executive, I want each Ready to Sale tag to show a Grade Price (min–max) next to the Cost Floor, so that I quote the right price quickly.

#	Given	When	Then
AC-01.1	Tag T: Laptop, grade B, a price row exists for (Laptop, B) with min ₹18,000 / max ₹21,000	I open `/sales/ready`	T shows Grade Price ₹18,000–21,000 and Cost Floor (today's `tag_min_price`)
AC-01.2	T has no grade, or no matching price row	I open the page	Grade Price shows "—" and a tooltip "No grade price for Laptop / "
AC-01.3	Grade min < Cost Floor	I open the page	Grade Price shows an amber warning icon "Below cost floor by ₹x"
AC-01.4	A more specific row exists (Laptop, B, make Dell, gen 8)	Price is resolved	The most specific match wins: make + generation > make > device type only

US-02 — Apply and override

As a Sales Executive, I want to apply grade prices to selected tags in one action and override any one with a reason.

#	Given	When	Then
AC-02.1	I select 40 graded tags	I click "Apply grade price"	`min_selling_price` / `max_selling_price` set from the grade row for each; tags with no row are skipped and listed in the result alert
AC-02.2	I set a manual price different from the grade price	I save	A reason is required; audit `TAG_PRICE_OVERRIDE` stores grade price, new price, reason
AC-02.3	A tag already has a manual price	Grade prices are re-applied	Manual price is kept unless I tick "Overwrite manual prices"

US-03 — Maintain the grade price list

As Managers/Admin, I want to maintain grade prices with effective dates and upload them by CSV.

#	Given	When	Then
AC-03.1	I add (Laptop, A, from 1 Jan 2027, min, max)	I save	Older row for (Laptop, A) is closed the day before; history kept
AC-03.2	I upload a CSV with a bad grade value	I submit	Count alerts: rows inserted vs rows with errors, with the row numbers

US-04 — Grading is consistent

As a Final QC Inspector, I want the grade list on the Final QC form to be the agreed sales grade scale, so that every grade has a price.

#	Given	When	Then
AC-04.1	The agreed scale is (example) A / B / C / D	I pass a tag	Only those values are offered; Grade is **required** on Pass

6. Functional requirements

ID	Requirement	Priority
FR-01	Grade price resolution service (server-side, `Decimal`): input device type (`sub_category`), grade, optional make, CPU generation; output min, max, source row id.	Must
FR-02	Ready to Sale Tag table: add Grade Price and Cost Floor columns and a below-floor warning.	Must
FR-03	"Apply grade price" bulk action and per-tag override with mandatory reason; audited.	Must
FR-04	Grade price list maintenance with effective dates and CSV upload (Managers/Admin).	Must
FR-05	One grade scale in Master Data used by Final QC form, Ready to Sale filters, CRM matrix. Grade required on Final QC Pass.	Must
FR-06	On Final QC Pass, store the resolved grade price as `expected_sale_value` (`models/engines.py::DeviceCosting`) so cost-engine scrap warnings can fire.	Should

ID	Requirement	Priority
FR-07	Report: margin by grade (sale price − device cost) per month.	Should
FR-08	As-Is lot suggested price from grade mix of its tags.	Could (phase 2)

7. Data and permission changes (Proposed)

Schema (dbdiagram approval required). Two options; recommendation is **Option A**.

Option	Change	Pros / cons
A	Extend `crm_grade_price_matrix` : add `min_sell` , `max_sell` , `cpu_generation` , `effective_from` , `effective_to`	Reuses an existing table and UI; CRM and refurbishment share one list
B	New `grade_sell_price` table	Clean separation; a second list to maintain

Also: `devices.grade_price_min` , `devices.grade_price_max` snapshot columns (nullable) written at Final QC pass and on Apply, so later list edits do not change history.

Audit actions: `GRADE_PRICE_APPLIED` (bulk), `TAG_PRICE_OVERRIDE` , `GRADE_PRICE_ROW_SAVED` .

Permissions: Maintain grade prices — Managers/Admin (the Price Matrix page permission in the Module Permission matrix; module key to confirm). Apply / override — roles that pass `_role_can_set_price` today. Pricing columns visible only where `can_view_pricing` is true.

8. UX notes

- Ready to Sale: two new Global Table columns after Unit Cost — "Grade Price" (`₹18,000–21,000`) and "Cost Floor". Below-floor uses `text-warning` icon with tooltip.
- Toolbar button "Apply grade price" next to "Bulk Set price"; result uses the multi-line `alert-info` (applied, skipped no-grade, skipped no-row, kept-manual).
- Override modal: price fields + required "Reason" textarea; errors per Design System §09.
- Grade badge colours: A success `#0F7A37` , B primary, C warning, D/E secondary.

9. Metrics

Metric	Baseline	Target	Source
Ready tags with a grade price	0%	100% of graded tags	`devices.grade_price_min`
Sales below grade min (count / month)	baseline: to be measured	to be agreed	Sale vs snapshot
Gross margin by grade	baseline: to be measured	to be agreed per grade	FR-07 report
Time from Ready to Sold (P6)	baseline: to be measured	−15% *(proposed)*	05 P6

10. Rollout

Phase	Scope	Gate
0	Agree grade scale (Open question 1); load first price list	Director + Sales lead sign
1	FR-01…05 (suggest only, no auto-apply)	UAT: SAL-01, FQC-01 + US-01…04
2	FR-06, FR-07; FR-08 As-Is	One month of override data reviewed

11. Open questions

#	Question	Owner
1	Which grade scale is the sales scale: A/B/C (Final QC form), A–E (`DeviceGrade`) or A/B/C/D/Mix (CRM)?	Final QC lead + Sales lead
2	Price key: device type + grade only, or also make, model, CPU generation, RAM / SSD?	Sales lead
3	Should Sales be able to sell below grade min without manager approval? (Today: warning only on below cost.)	Director
4	Who updates the list, and how often?	Sales Manager

PART D

PRD-04 · Invoice & E-way Bill from Sale

Field	Value
Status	Draft
Roadmap item	X4 phase 1 (Next, Q1 2027) and L2 phase 2 (Later, Q2 2027) — 02_Roadmap.md
Business owner	Accounts head (with Sales lead)
Personas	Sales Executive, Accounts, Managers/Admin, gate staff (Gate Pass)
Related	Project skill `server-side-financial-snapshot`; `docs/product-architecture/02-architecture/05_Integration_Specs.md` (planned in the folder README); Vision open question Q4 (e-invoice)

This PRD does not give tax advice. Every GST rule, HSN code and threshold named below must be confirmed by Accounts before build.

1. Problem

A sale in the app records tags, price, customer and an invoice number that someone types in or reads from an uploaded PDF. The app can print a tax invoice and a delivery challan, but **one per tag**: a 40-laptop sale prints 40 single-line invoices. The invoice number has no series or check for gaps. The e-way bill — needed to move goods above the statutory value — is captured only for goods coming **in** (lots, GRN, Parts GRN), never for goods going **out**. Accounts re-keys the sale into the billing tool to make the real invoice and the e-way bill, and the gate lets goods out without checking either.

2. Personas

Persona	Need
Sales Executive	Finish a sale and print one correct invoice and challan for the whole order
Accounts	A gap-free invoice series per company and FY, correct GST split, e-way bill recorded against the invoice
Gate staff	See that the invoice and (when needed) e-way bill exist before confirming exit
Managers/Admin	Sales list and P&L that match invoices

3. Goals and non-goals

Goals	Non-goals
One invoice document per sale event, with all tags as lines	Replacing the accounting software / GL

Goals	Non-goals
Server-generated invoice number series per selling company and FY	Credit-note document redesign (separate flow exists)
E-way bill number, date, vehicle, transporter captured on the invoice; printed on challan and gate pass	Inbound e-way bill changes (already captured on lots / GRN)
Warn (configurable: block) gate exit when e-way bill is required but missing	Phase 1 does not call any government API
Phase 2: generate e-way bill (and e-invoice IRN, if required) through a GST provider	

4. Current behaviour (from code)

Area	What happens today	Source
Sale creation	One form; `barcode` may be a comma list or `price_groups` JSON; **one `Sale` row per tag**; invoice number, customer, state, address, payment mode shared across the rows	`routers/sales.py::create_sale`
Sale number	From PostgreSQL sequence `sale_number_seq` (self-heals if missing)	`routers/sales.py::ensure_sale_number_seq`
Invoice number	Free text `Sale.invoice_no`, optional; can be auto-filled by parsing an uploaded PDF (`pdfplumber`, best effort). On sale, the tag's `Device.invoice_number` is replaced by the sale invoice number.	`routers/sales.py::parse_invoice_pdf`, `create_sale`
Company snapshot	Selling company name, GSTIN, address, state, state code, phone, email frozen onto each Sale	`models/sales.py::Sale` (`company_*`), `routers/invoices.py::_sale_company_snapshot`
Tax invoice print	`/invoices/print/{sale_id}` — one Sale = one device = one line. GST back-calculated from the inclusive sale price; intra-state (CGST + SGST) when customer state equals company state, else IGST. Rates from CostConfig `gst_rate_intra` / `gst_rate_inter`, default 18%. **Customer state defaults to "Delhi" when blank.** HSN printed as **8471 for every device** (TFT monitors included). Invoice no. shown = `invoice_no` or `sale_number`.	`routers/invoices.py::print_invoice`, `_compute_gst`, `_fetch_gst_rates`, `templates/invoices/print.html`
Delivery challan	`/invoices/waybill/{sale_id}` titled "Delivery Challan", one per Sale	`routers/invoices.py::print_waybill`, `templates/invoices/waybill.html`
Transport	Transport mode, courier, tracking no., dispatch date, delivery status on Sale	`routers/sales.py::sale_update_transport`
Gate pass	Exit confirm (single / bulk) sets `delivery_status = dispatched`, dispatch date; audited `GATE_EXIT_CONFIRMED`; no invoice / e-way bill check	`routers/gate_pass.py::confirm_exit`, `bulk-exit`
E-way bill (inbound only)	`Lot.e_way_bill`; GRN form field; `PartsGRN.eway_bill_number`, `eway_bill_date`, `eway_bill_file`	`models/lot.py`, `routers/grn.py`, `models/parts_grn.py`, `routers/parts_grn.py`
E-way bill (outbound)	**None.** No field on `sales`.	`models/sales.py`

Area	What happens today	Source
Accounts	Customer receipts (single / bulk CSV), supplier payments, PO / quote invoice upload	`routers/accounts.py`

5. User stories and acceptance criteria

US-01 — One invoice per sale

As a Sales Executive, I want one invoice for all tags sold together, so that the customer gets one correct document.

#	Given	When	Then
AC-01.1	I sell 3 laptops at ₹20,000 and 2 monitors at ₹6,000 in one submit	The sale saves	One invoice header is created with 5 lines (or 2 grouped lines by model / price — see Open question 3), totals computed server-side in `Decimal`
AC-01.2	The invoice exists	I open Print Invoice from any of the 5 Sale rows	The same single invoice opens
AC-01.3	Customer state is blank	I try to save	Save is refused: "Customer state is required for GST" (no silent "Delhi")

US-02 — Invoice number series

As Accounts, I want the app to issue invoice numbers from a series per selling company and FY, so that there are no gaps or duplicates.

#	Given	When	Then
AC-02.1	Series for "OxyPC Computers", FY 2026-27, prefix configured	A sale saves	Invoice number = next in series (e.g. `<prefix>/26-27/00042`), unique per company + FY
AC-02.2	Two sales save at the same moment	—	Different numbers, no gap (DB sequence / row lock per series)
AC-02.3	An invoice must be cancelled	Accounts cancels it with a reason	Number is kept as "Cancelled" (not reused); audit logged; linked tags handled by the Return / Credit Note flow
AC-02.4	Legacy flow (typed invoice number) is still enabled	Sales types a number	Allowed only when setting `invoice_numbering = manual` ; duplicates per company + FY refused

US-03 — Record the e-way bill (phase 1)

As Accounts, I want to record the e-way bill on the invoice, so that it prints on the challan and the gate can check it.

#	Given	When	Then
AC-03.1	Invoice value is above the configured threshold	I open the invoice	A banner says "E-way bill required" until number + date are saved
AC-03.2	I enter EWB number, date, vehicle number, transporter	I save	Values stored, audited `EWAY_BILL_RECORDED` , printed on Delivery Challan and Gate Pass

#	Given	When	Then
AC-03.3	Value is below the threshold	—	No banner; fields optional

US-04 — Gate checks before exit

As gate staff, I want the gate pass to show invoice and e-way bill status, so that goods do not leave without papers.

#	Given	When	Then
AC-04.1	E-way bill required but missing; setting `gate_requires_ewb = warn`	I confirm exit	Exit allowed with a warning; audit notes "exit without EWB"
AC-04.2	Same, setting `gate_requires_ewb = block`	I confirm exit	Refused: "E-way bill missing for invoice "
AC-04.3	Bulk exit of 10 sales, 2 missing EWB, block mode	I submit	8 exit, 2 listed as skipped in the result alert

US-05 — Generate e-way bill from the app (phase 2)

As Accounts, I want to generate the e-way bill from the invoice, so that I do not re-key it on the government portal.

#	Given	When	Then
AC-05.1	GST provider credentials configured	I click Generate EWB	Request built from invoice data; returned number / date / validity stored; failure message shown verbatim
AC-05.2	E-invoice (IRN) is required for the company (Accounts decision)	Invoice is finalised	IRN and QR obtained first, then EWB

6. Functional requirements

ID	Requirement	Priority
FR-01	Create a sale invoice header for each sale submit; link every Sale row of that submit to it.	Must
FR-02	Print invoice from the header: all lines, per-line HSN by category (table in Master Data), taxable value, CGST / SGST / IGST, totals; company and customer snapshot (reuse `_sale_company_snapshot`).	Must
FR-03	Server-generated invoice number per (company, FY) series; prefix and start number set by Admin; cancellation keeps the number.	Must
FR-04	Customer state (and GSTIN for B2B) required; remove the "Delhi" default in `_compute_gst`.	Must
FR-05	E-way bill fields on the invoice header (number, date, valid-until, vehicle no., transporter, transporter ID, distance) with threshold rule in `AppSetting`.	Must
FR-06	Delivery Challan and Gate Pass print from the header and show EWB.	Must
FR-07	Gate exit check with setting off / warn / block.	Should

ID	Requirement	Priority
FR-08	Backfill: group existing Sale rows by (`invoice_no` , `company_id` , `sold_at` date, customer) into headers; leave ungroupable rows as single-line headers.	Should
FR-09	Sales list and exports show invoice header number and EWB status.	Should
FR-10	Phase 2: GST provider integration for EWB (and IRN if required), with retry and error log.	Could (phase 2)

7. Data and permission changes (Proposed)

Schema (dbdiagram approval required; follow `server-side-financial-snapshot` — totals stored as columns, computed in `Decimal`):

Table	Columns	Notes
`sale_invoices` (new)	`id` , `invoice_number` (unique per company + FY), `company_id` + snapshot columns, `fy` , `invoice_date` , customer snapshot (name, GSTIN, state, state code, address), `taxable_total` , `cgst` , `sgst` , `igst` , `grand_total` , `status` (issued / cancelled), `cancel_reason` , `ewb_number` , `ewb_date` , `ewb_valid_until` , `vehicle_no` , `transporter_name` , `transporter_id` , `irn` (nullable), `created_by` , `created_at`	Header
`sales`	add `sale_invoice_id` FK (nullable), `hsn_code` , `taxable_value` , `tax_rate`	Each Sale row becomes an invoice line
`invoice_series` (new)	`company_id` , `fy` , `prefix` , `next_number`	Row-locked on issue
Master Data	`hsn_by_category` (Laptop, Desktop, TFT Monitor, ...)	Values confirmed by Accounts
`AppSetting`	`invoice_numbering` (auto / manual), `ewb_threshold_amount` , `gate_requires_ewb` (off / warn / block)	

Audit actions: `INVOICE_ISSUED` , `INVOICE_CANCELLED` , `EWAY_BILL_RECORDED` , `EWAY_BILL_GENERATED` , `GATE_EXIT_WITHOUT_EWB` .

Permissions: Issue invoice — whoever can create a sale today (`require_module_perm("sales", "add")`). Cancel invoice, edit series, record / generate EWB — Accounts (custom role via matrix) and Admin. Gate exit unchanged.

8. UX notes

- After Save Sale: success alert (`#0F7A37`) "Sale saved · Invoice · 5 tags" with buttons Print Invoice, Print Challan.
- Sale Detail: an "Invoice" card (Global Table style) listing lines; EWB status badge: Not needed (secondary) / Required (warning) / Recorded (success).
- EWB modal: fields in Design System form layout; errors keep values (§09).
- Gate Pass Global Table: new columns Invoice No. and EWB; blocked rows show a red badge and are excluded from Bulk Exit with the standard multi-line result alert.
- Print templates stay A4, one invoice may run to several pages; repeat header on each page.

9. Metrics

Metric	Baseline	Target	Source
Sales with an invoice number	baseline: to be measured (`sales.invoice_no` null rate)	100%	`sale_invoices`
Invoices per multi-tag sale	n tags = n invoices today	1	`sale_invoices`
Outbound consignments above threshold with EWB	0% recorded in app	100%	`sale_invoices.ewb_number`
Gate exits without EWB (warn mode)	baseline: to be measured	0 after 1 month	audit `GATE_EXIT_WITHOUT_EWB`
Accounts re-keying time per sale	baseline: to be measured (time study)	−70% *(proposed)*	time study

10. Rollout

Phase	Scope	Gate
1a (Q1 2027)	FR-01…04, FR-08 backfill; `invoice_numbering = manual` first, then switch to auto at FY or month start	Accounts reconciles one month of invoices vs billing tool on staging
1b (Q1 2027)	FR-05…07, FR-09; gate in `warn` mode	UAT ACC-03, SAL-06 + US-03/04
2 (Q2 2027)	FR-10 GST provider integration; gate `block` mode	Provider contract; Accounts sign-off

11. Open questions

#	Question	Owner
1	Is e-invoice (IRN) mandatory for each selling company / entity (the dashboard entity list shows OxyPC Computers, Renew Circuits, Deshwal)?	Accounts
2	E-way bill threshold and rules for intra-state movement — confirm value and any state-specific rule.	Accounts
3	Invoice lines: one per tag (serial on line) or grouped by model and price with tag list annexure?	Accounts + Sales lead
4	HSN per category — the current template prints 8471 for all devices; confirm correct codes for monitors and others.	Accounts
5	Which GST provider / GSP, and who holds credentials?	Accounts
6	Do replacement sales (₹0, "Replaced with …") need a delivery challan only, not an invoice?	Accounts

PART E

PRD Template

Field	Value
Status	Draft / In review / Approved / Built / Shipped
Roadmap item	02_Roadmap.md>
Business owner	
Personas	
Related	/qa/requirements)>

Rule: describe **current behaviour only from code** and cite `file::function`. Mark everything new as **Proposed**.

1. Problem

What hurts, for whom, how often. One short paragraph and the evidence (screens, counts, code).

2. Personas

Persona	What they need from this feature

3. Goals and non-goals

Goals	Non-goals

4. Current behaviour (from code)

Area	What happens today	Source
		`path/file.py::function`

5. User stories and acceptance criteria

US-01 —